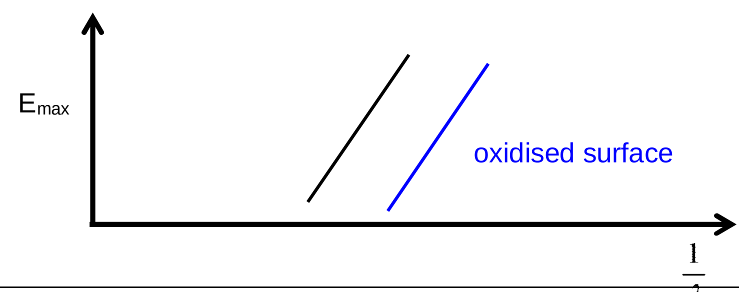
	intensity of light at the bright fringe $= \dfrac{\quad}{(4.0)^2} = 0.49$
5(a)	By conservation of energy, Energy of incoming photon = Work done to remove electron from metal + Kinetic energy of the most energetic photoelectron $hf = \Phi + E_{max}$ $E_{max} = h\dfrac{c}{\lambda} - h\dfrac{c}{\lambda_0} = hc\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda_0}\right)$
5(b)(i)	The maximum wavelength at which emission of electrons occurs when $E_{max} = 0$. Extend the graph in Fig 5.1 till it intersects the $\dfrac{1}{\lambda}$ axis. At the intersection point, $\dfrac{1}{\lambda_0} = 2.3 \times 10^6$ $\lambda_0 = 4.3 \times 10^{-7}$ m
5(b)(ii)	Gradient of the graph in Fig 5.1 is given by hc. $hc = \dfrac{1.0 \times 10^{-19}}{0.5 \times 10^6}$ $h = \dfrac{1.0 \times 10^{-19}}{0.5 \times 10^6 \times 3.0 \times 10^8} = 6.67 \times 10^{-34}$ Js
5 (c)	The new graph will have the same gradient as the original graph in Fig 5.1. Since the work function energy of the oxidised surface has increased, this implies that the threshold frequency (ie $\dfrac{c}{\lambda_0}$) will increase. Hence, the new graph will be shifted to the right as shown in sketch below. 
6(a)(i)	It is a region of space where a non-contact force acts on a particle by virtue of